Dynamic LION*

*LION: Empowering Multimodal Large Language Model with Dual-Level Visual Knowledge

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***Abstract*—In the past few years, Multimodal Large Language Models (MLLMs) have shown strong potential in expanding the modalities that the LLMs can process. The performance of LLMs with multimodal data continues to improve by leveraging the knowledge between modalities. One recent work, the dual-*L*evel v*I*sual kn*O*wledge e*N*hanced Multimodal Large Language Model (LION), demonstrates improvements over existing MLLMs in visual knowledge. LION incorporates visual knowledge through the progressive integration of fine-grained spatial-aware features and the use of soft prompting with high-level semantic visual evidence. Despite these advancements, the soft prompting technique remains underexplored. As mentioned from the LION paper, soft prompting method still has the danger from hallucination to some degree. In this report, we propose a dynamic soft prompting to mitigate the object hallucination problem, and explore its effectiveness.**

***Index Terms*—Multimodal Large Language Model (MLLM), LLM, Vision Encoder**

I. Introduction

Recent research shows an increasing use of Multimodal Large Language Models (MLLMs) with its ability to incorporate different types of knowledge across various modalities (*i.e.*, image and text) [1]. This is especially useful for vision language (VL) tasks, such as reasoning of the input image using natural language style knowledge. Nevertheless, there still exists limitations of extracting and reasoning of visual knowledge. This is largely due to most of the existing MLLMs employ vision encoder that was pretrained on a coarsely aligned image-text pairs, which ultimately leads to insufficient extraction of visual information.

One recent work, dual-**L**evel v**I**sual kn**O**wledge e**N**hanced Multimodal Large Language Model (LION) [2], mitigates the issue by progressively incorporates fine-grained spatial-aware visual knowledge and applies soft prompting of high-level semantic visual envidence. Specifically, the authors propose a stage-wise instruction-tuning strategy to perform image-level and region-level VL tasks. It learns the visual knowledge separately by using two different adapters, and then a router module is used to aggregate the knowledge from image-level and region-level adapters into a single vision knowledge representation. In addition, LION employs a very well-known foundation model, Recognize Anything Model (RAM) [3], as a vision extractor to obtain image tags off-the-shelf. Tags extracted from RAM are used to generate a soft prompt, which is the key component of the paper. A soft prompt is an instruction template with image tags and trainable parameters representing the prompt. The authors suggest the template as such, *"According to <hint>, you are allowed to use the following tags:"*, where the <*hint*> will be replaced with a trainable soft prompt, and the tags from RAM will be inserted at the end of the template. In this way, the model can avoid potential negative influence by the imperfect predictions from RAM.

II. Limitations of the Previous Work

While LION achieves high performance in image captioning, visual question answering (VQA), and visual grounding tasks, object hallucination is one of the most concerned issue from the paper, as the authors mentioned. This challenge is due to an imperfect vision model such as RAM,

To be more specific on soft prompting, the model employs RAM to extract image tags in order to generate soft prompt. However, since vision models such as RAM are often imperfect, potential object hallucination problem is inevitable. Not only does the performance of off-the-shelf vision models contribute to the problem, but over-reliance on statistical bias [4] and unimodal priors [5], [6] can also lead to challenges such as visual hallucinations. Nevertheless, foundation models in MLLMs have shown impressive performance with ease of use [7].

soft prompting technique is still underexplored. In order to avoid object hallucinations while maintaining foundation models,

Not only that, the soft prompting technique from the paper uses a simple image tag information,

III. Related Work

W. Fang *et al.* [8] Y. Zeng *et al.* [9] fine-grained contrastive learning

A. Soft Prompting

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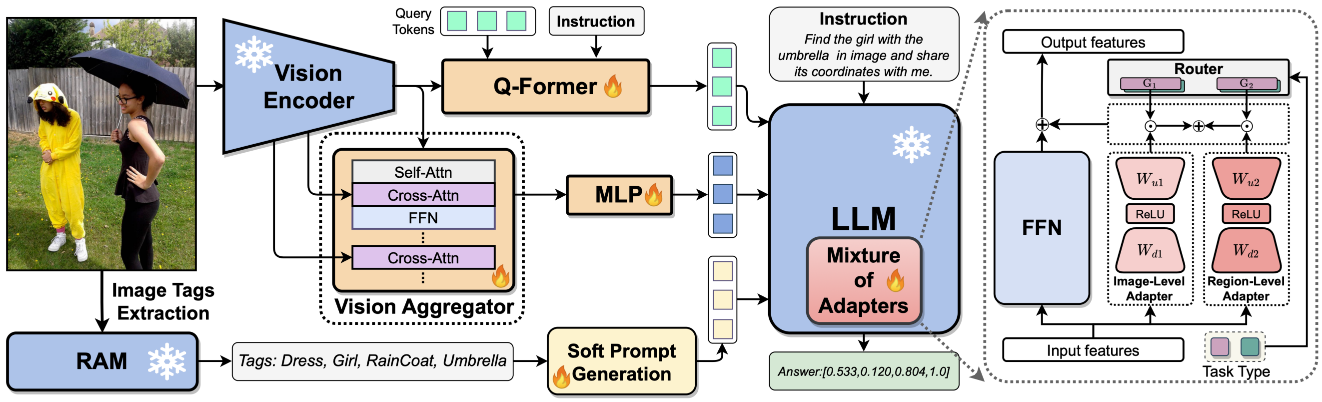

Fig. 1. Overview of the original LION [2]. The model extracts visual features from Q-Former, and combines them with fine-grained spatial-aware features from the vision aggregator. The Mixture-of-Adapters with a router in the frozen LLM dynamically fuses visual knowledge learned from different visual branches and LLM adapters based on the task types (image-level and region-level).

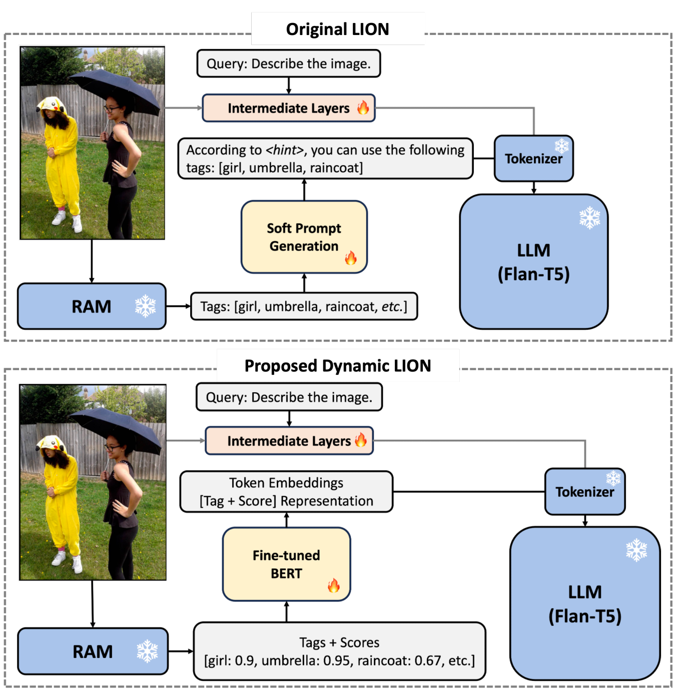

Fig. 2. Comparison between the original LION's soft prompting and the proposed dynamic soft prompting.

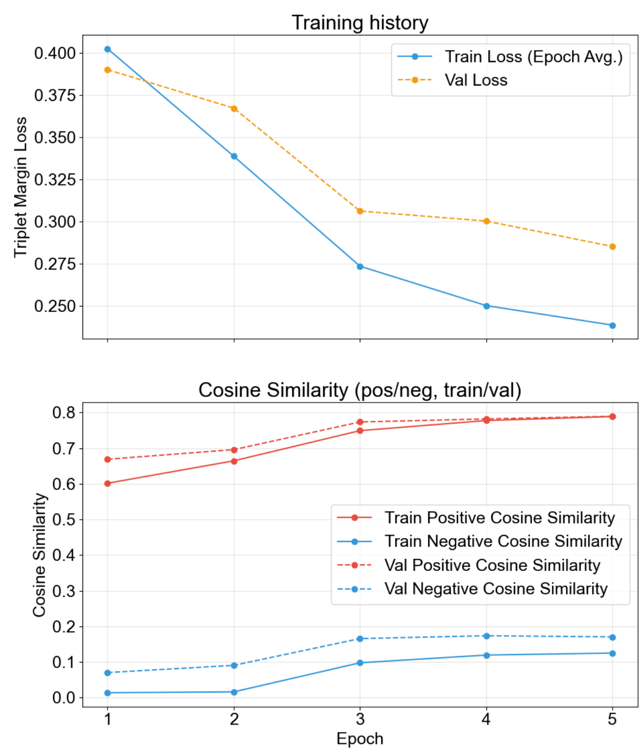

Fig. 3. Fine-tuning result of BERT model aligned to token embeddings.

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IV. Proposed Method

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A. Model Architecture

V. Experiments

A. Fine-tuning BERT

B. Re-training the LION with Dynamic Soft Prompt

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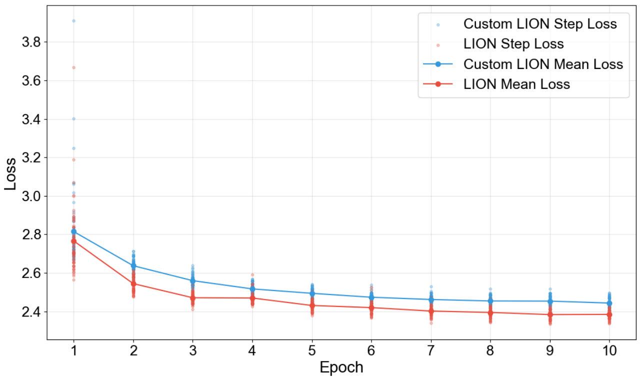

Fig. 4. Training results of original LION model and custom LION model (*i.e.*, dynamic soft prompting).

Model	COCOCap	TextCaps	OKVQA	AOKVQA
LION-small	69.09	26.45	0	0
LION-small (ours)	0	0	0	0
LION-4B	138.20	104.87	51.08	59.98
LION-12B	139.25	108.76	57.33	60.87

TABLE I
COMPARISON ON IMAGE CAPTIONING AND VQA.

C. *Evaluations on Image-Level VL Tasks*

We evaluate the dynamic soft prompting method's performance on two types of image-level VL tasks, which are image captioning and VQA. For image captioning, COCO caption [10] and TextCaps [11] datasets are used, and report CIDEr as the evaluation metric. Also for VQA task, OKVQA [12] and AOKVQA [13] datasets are used, and its performance is measured by Top-1 Accuracy metric.

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